

26

THE NUMBER OF INSULATED WIRES IN LESAGE'S TELEGRAPH

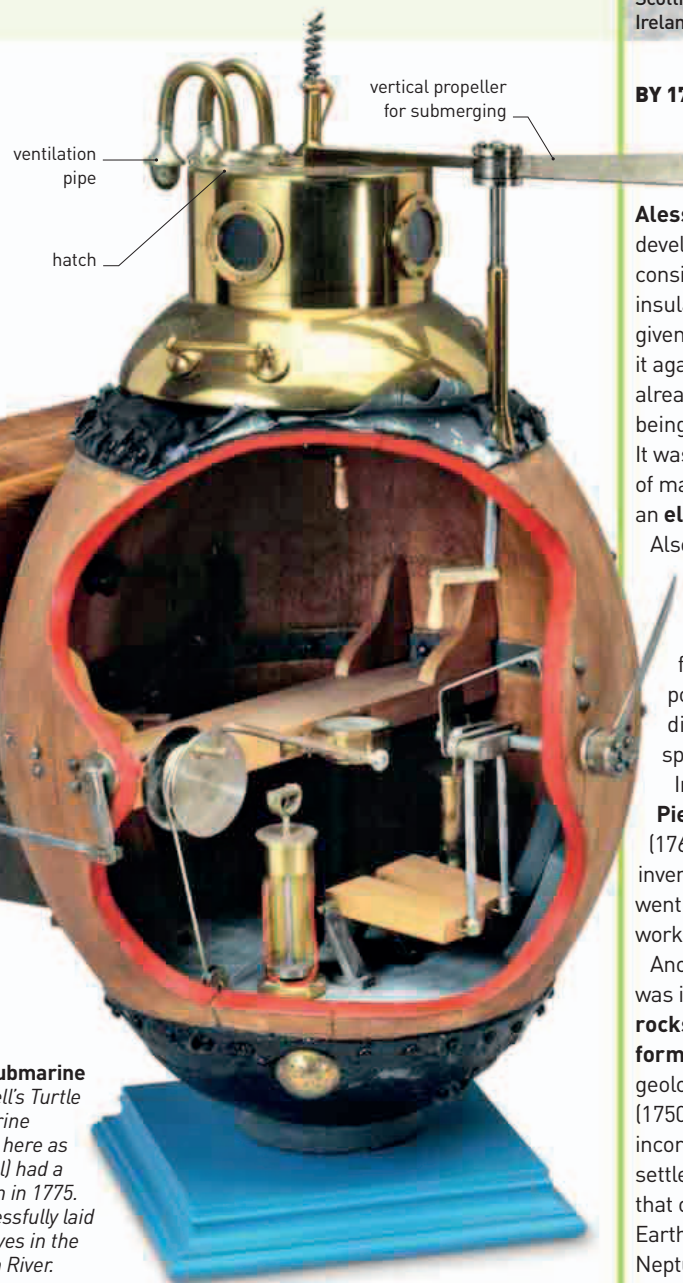


mercuric oxide and various nitrates in 1772. French chemist **Antoine Lavoisier** (1743–94) later claimed to have first discovered oxygen and gave it its name—but both Priestley and Scheele had already told him about their discoveries.

American inventor **David Bushnell** (1742–1824) built an underwater explosive device in 1773 to help the American Revolutionary War effort. In 1775, he held trials of the vessel intended to deliver this device, the world's first working submarine. Known as the Turtle, but shaped like a lemon, it had an airtight hatch at the top for the submariner and was maneuvered by hand-cranked propellers and rudder.

The following year, **Carl Wilhelm Scheele** discovered a gas for which he is credited—**chlorine**. He called his discovery “dephlogisticated muriatic acid air” because it came from muriatic acid, now known as hydrochloric acid.

In Geneva, Switzerland, in 1774, physicist **Georges-Louis LeSage** (1724–1803) created an early form of electric telegraph. It had a separate wire for each of the 26 letters of the alphabet and could send messages between rooms.



First submarine
Bushnell's Turtle submarine (shown here as a model) had a trial run in 1775. It successfully laid explosives in the Hudson River.



Scottish chemist James Keir realized that the strange columns of Northern Ireland's Giant's Causeway are made from molten lava.

BY 1775, ELECTRICITY was the subject of numerous experiments and this year Italian physicist **Alessandro Volta** (1745–1827) developed the **electrophorus**. It consisted of a metal disk with an insulating handle that could be given a **static charge** by holding it against a disk of **resin** that had already been given a charge by being rubbed with fur or wool. It was a simple and effective way of magnifying and accumulating an **electric charge**.

Also this year, Swedish naturalist **Peter Forsskål's** (1732–63) studies of Middle Eastern fauna were published posthumously; Forsskål had died of malaria while specimen hunting in Yemen.

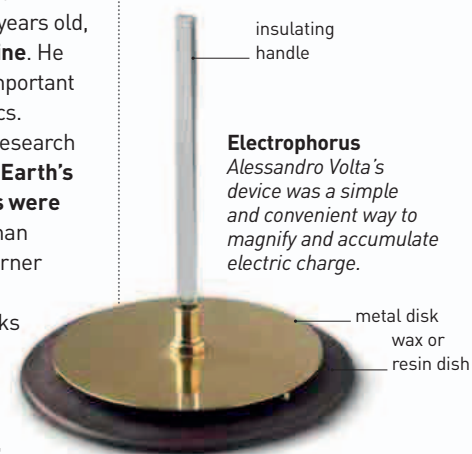
In France, future engineer **Pierre-Simon Girard** (1765–1836), just ten years old, invented a **water turbine**. He went on to complete important work on fluid mechanics.

Another key area of research was investigating **how Earth's rocks and landscapes were formed**. In 1776, German geologist Abraham Werner (1750–1817) insisted, incorrectly, that all rocks settled out of a ocean that once covered the Earth—a theory called Neptunism. Plutonists,

40,000

THE NUMBER OF BASALT COLUMNS THAT MAKE UP THE GIANT'S CAUSEWAY

however, believed that rocks were formed by **volcanic processes**. Plutonist **James Keir** (1735–1820), a Scottish chemist, realized that the interlocking basalt columns of formations like the Giant's Causeway in Northern Ireland were formed by the **crystallization of molten lava**.



Electrophorus
Alessandro Volta's device was a simple and convenient way to magnify and accumulate electric charge.

1773 American inventor **David Bushnell** builds an explosive device

1774 Swedish chemist **Carl Wilhelm Scheele** discovers chlorine, manganese, and barium

1774 Swiss physicist **Georges-Louis LeSage** creates the first electric telegraph

August 1, 1774
English chemist **Joseph Priestley** isolates oxygen

1775 Swedish naturalist **Peter Forsskål's** observations on Middle Eastern fauna are published

1775 French future engineer **Pierre-Simon Girard**, aged 10 years old, invents a water turbine

1775 Bushnell completes the Turtle submarine

1775 Italian physicist **Alessandro Volta** invents the electrophorus

1776 Scottish chemist **James Keir** suggests some rocks are crystallized lava